

USER VOICE INPUT



Wake Word Detection
(‘Hey SentinAL’, local)



Speech-to-Text (STT)
Whisper local / Azure STT



NLP Correction Layer
hallucination/refusal filter



HYBRID INTENT ROUTER (see Figure 2 for full decision logic)

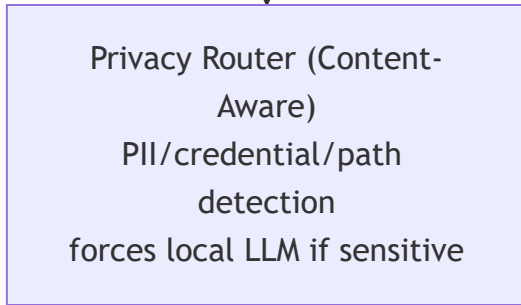
Tier 1: Deterministic
Keyword
Fast-Path (match -> bypass
LLM)



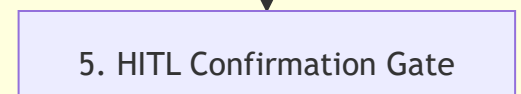
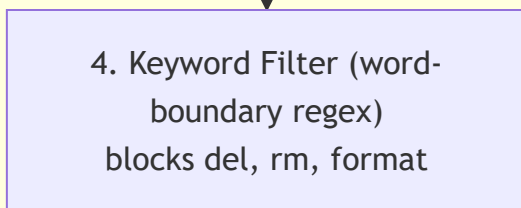
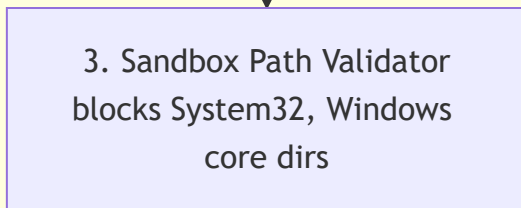
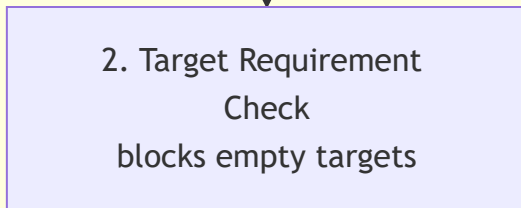
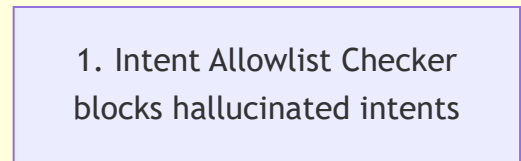
Tier 2: Semantic
Embedding
(all-MiniLM-L6-v2, conf >
0.40 -> map intent)



Tier 3: LLM Extraction
Fallback
(cloud/local, per Privacy
Router)



VALIDATION PIPELINE (see Figure 4 for full sequence)



user consent for deletions

Execute Pipeline
capability dispatch
(23 registered capabilities)

Postcondition Observer
Process -> Window -> VLM
verification

TTS Response
spoken feedback to user

